StreamScale

Kafka → transactional PostgreSQL · scaling experiment

27 measured runs · 3 modes × 3 scenarios × 3 repetitions

All acknowledged IDs reconciled before/after restart

Same image, logical scenario sequence, resources and partitions

30-second warmup + 60-second base window; late fixtures extend the tail

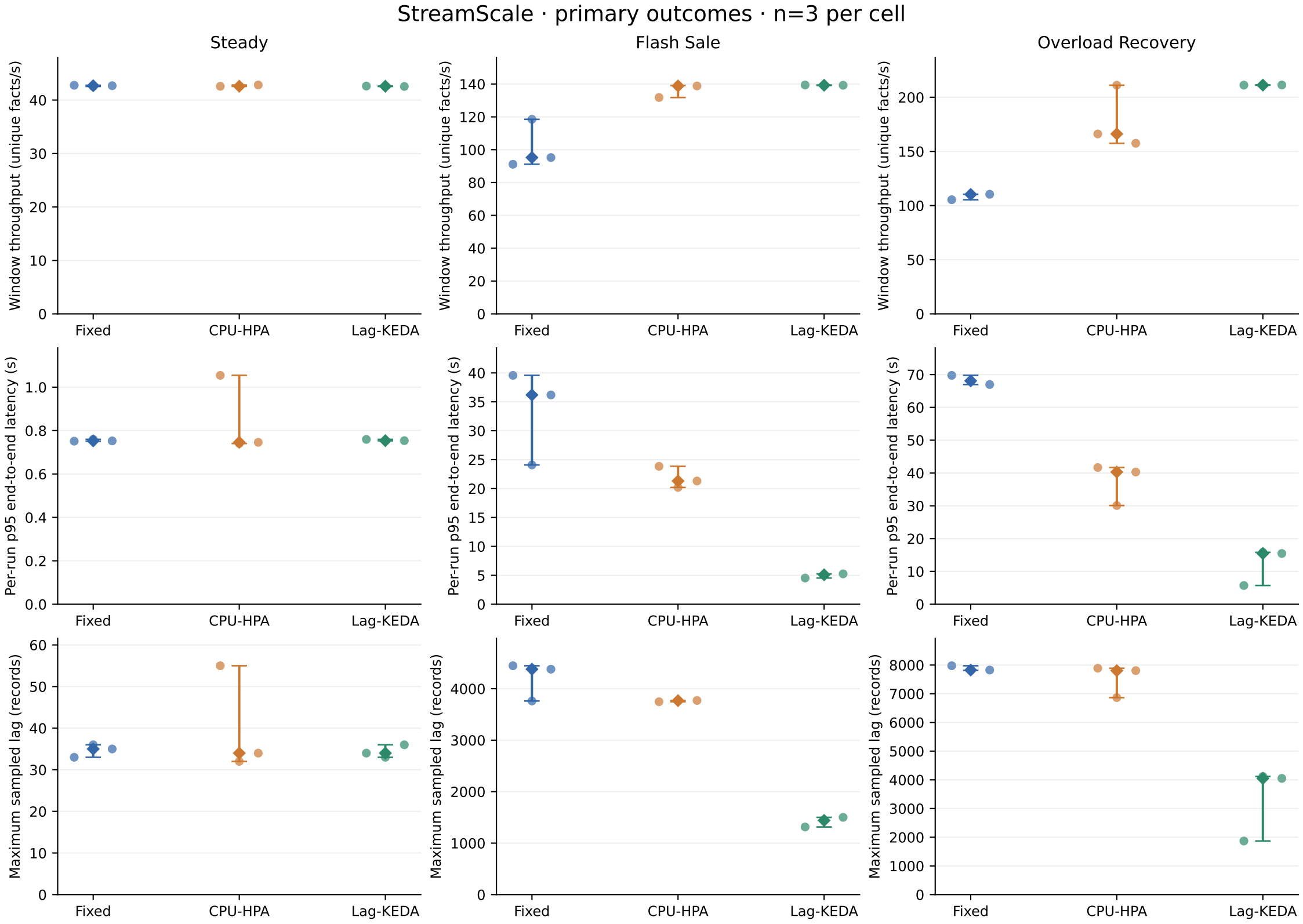
Frozen rates: 44/s steady · 336/s surge

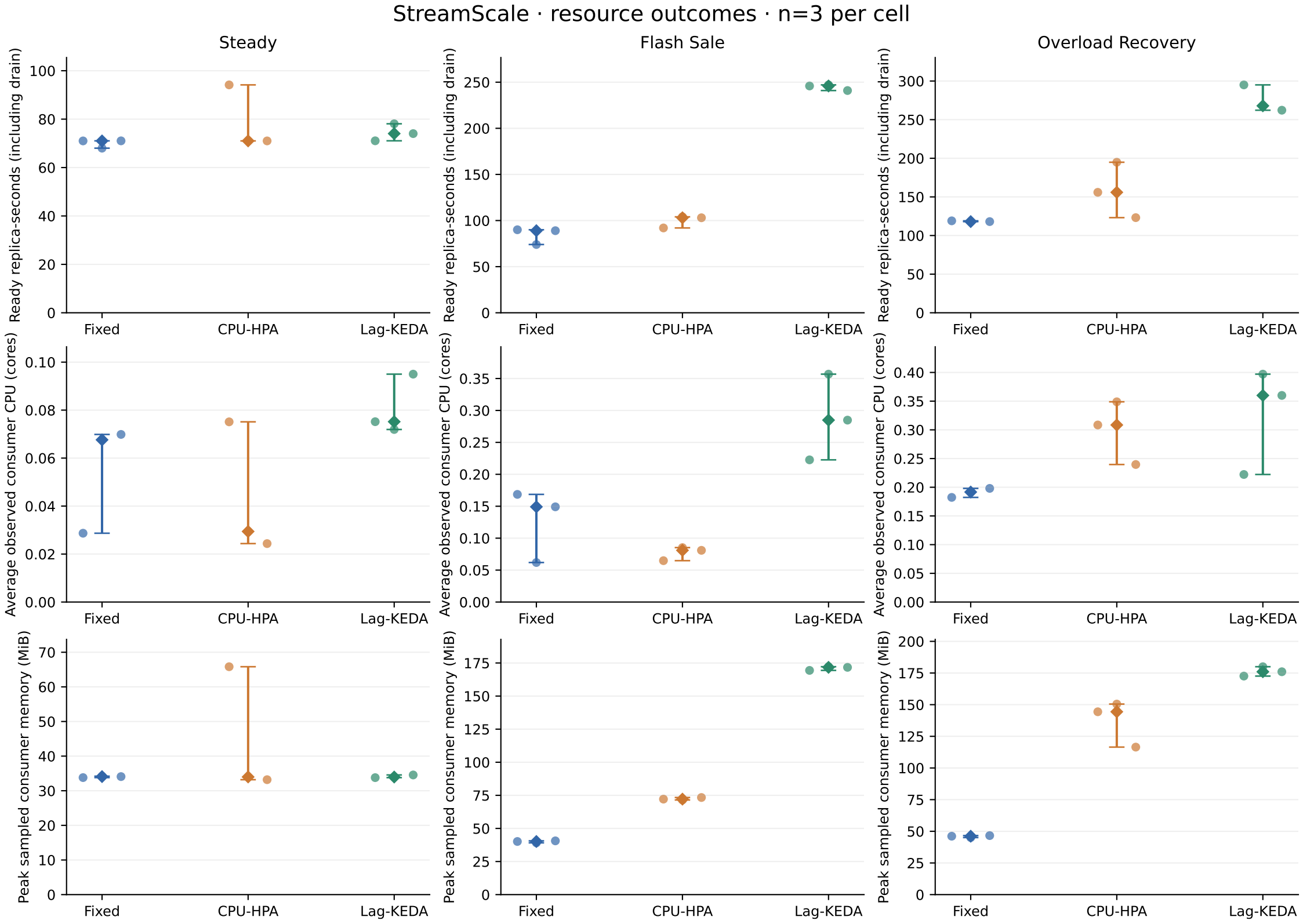
Dots = repetitions; diamonds = median; whiskers = observed range

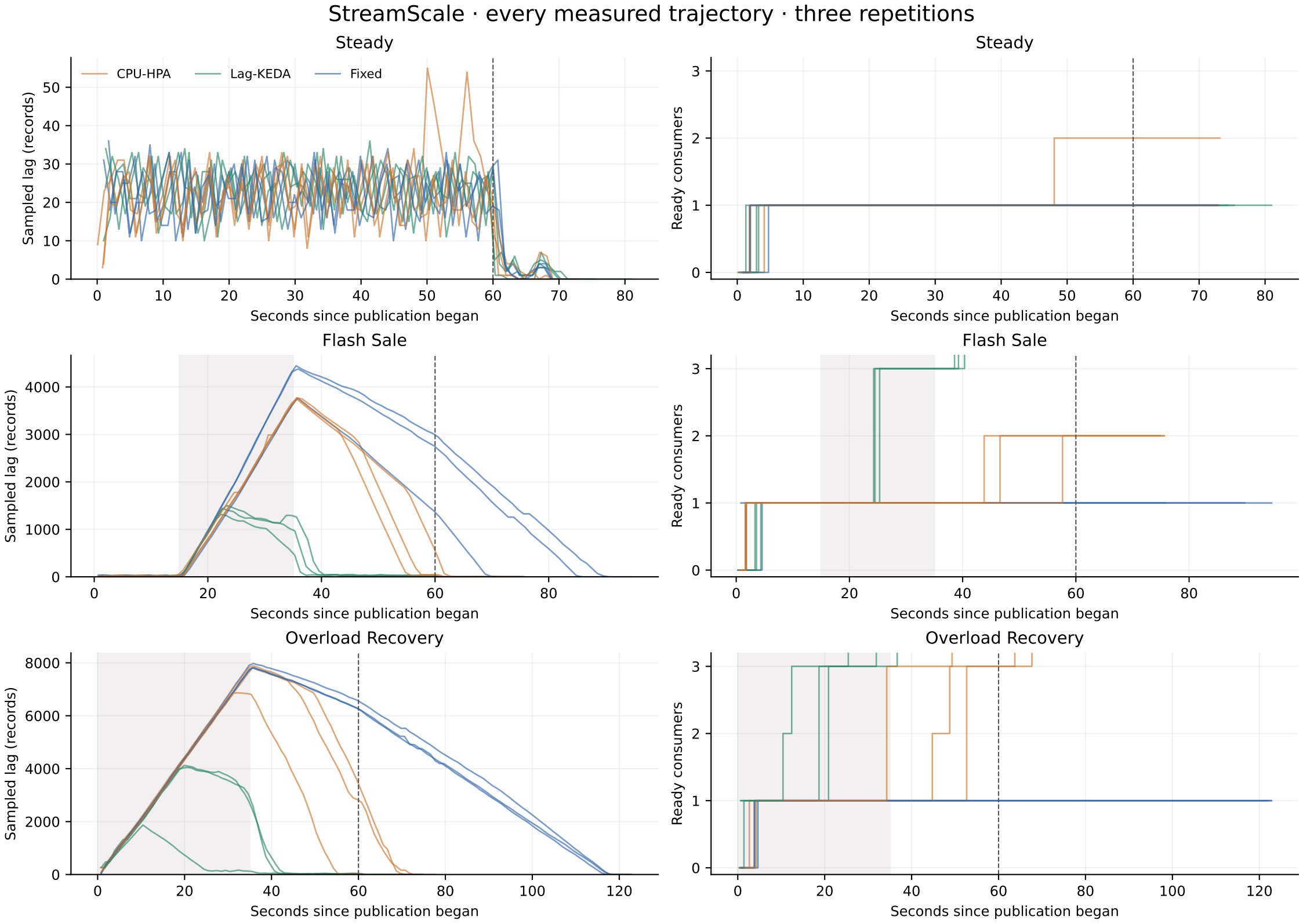
SQL p95 includes the drained tail; throughput uses the publishing window

One shared kind node, small sample, sampled resources; descriptive results

See report.md, summary.csv, frozen-protocol.json and provenance.json


StreamScale · primary outcomes · n=3 per cell
Steady
Flash Sale
Overload Recovery
Window throughput (unique facts/s)
Per-run p95 end-to-end latency (s)
Maximum sampled lag (records)
Fixed
CPU-HPA
Lag-KEDA


StreamScale · resource outcomes · n=3 per cell
Steady
Flash Sale
Overload Recovery
Ready replica-seconds (including drain)
Average observed consumer CPU (cores)
Peak sampled consumer memory (MiB)
Fixed
CPU-HPA
Lag-KEDA


StreamScale · every measured trajectory · three repetitions
Steady
CPU-HPA
Lag-KEDA
Fixed
Sampled lag (records)
Ready consumers
Seconds since publication began
Flash Sale
Overload Recovery